



Group Manage Meeting Minutes

Location	A402
Date	13/10/2021
Time	8:40 ~ 10:10
Attendance	Wu Ziyue ; Zhao Chenyu ; He Liusiyuan ; Liu Zhening ; Peng Xuelin ; Xu Xiuqi
Chair	Liu Zhening
Secretary	Zhao Chenyu

Minutes :

1. Define the problem
 - 1) How to use limited materials to make the bridge strong
 - 2) How to make the unfixed bridge stable and not fall from the support
2. Issue raised or to be discussed
For this part we list six ideas: Arch bridge; Suspension bridge; Beam bridge; Truss bridge; Triangular bridge; The 'special' bridge
3. Action plan

Who	Task	Format	DDL
Peng Xuelin	Identify the problem	File	13/10/2021
Wu Ziyue	Record the meeting	File	13/10/2021
Liu Zhening	The chairman and combine the documents and submit on time.	File	13/10/2021
Xu Xiuqi	Brainstorm possible solutions and evaluate the ideas	File	13/10/2021
He Liusiyuan	Brainstorm possible solutions and evaluate the ideas	File	13/10/2021
Zhao Chenyu	Record the meeting	File	13/10/2021

4. Date of next meeting: Oct.15th 10:30-12:00

Seven Step Plan:

Step1: Minimize deflection: This means higher stiffness. And the structure can make the 1kg load have a minimal effect.

Step2: The problem is to use limited materials to make our bridge as strong as possible. Since both ends cannot be fixed and the middle of the bridge deck cannot be contact with the ground, our solution is to adopt a different special structure to make the bridge deck obtain higher strength.

Step3: After the brainstorm discussion. There are six ways to build the bridge: the arch bridge, the suspension bridge, the beam bridge, the truss bridge, the triangle bridge, and the "Special" bridge.

Step4: Focusing on these six bridge building ways, there are some detailed specific production processes:



Arch bridge: Adhere the straws end to end with the adhesive tape to build the fundamental basis of the bridge. Bend another group of straws at a slight angle to build the arching part. Use the adhesive tape to connect two parts of the bridge to keep the bending angle.

Suspension bridge: Erect two groups of adherent straws as two walls of the bridge. Bundle five or more straws together and adhere them end to end as the deck. Connect the deck at the top of two straw walls.

Beam bridge: Use the adhesive tape to adhere the paper drinking straws side by side one by one to form the main body of the beam bridge. Divide the rest straws into 2 stacks. Using adhesive tapes to glue the two stacks of paper straws into two bundles, which forms the two abutments of the bridge. Link the bridge body and the two abutments to form the beam bridge.

Truss bridge: Use the adhesive tape to stick the straws end to end to build the bottom and upper bottom of the trapezoidal truss structure. And use the rest straws as the top chords and bottom chords of the trapezoidal truss structure. Fix each part of the truss structure by tape and strengthen the bridge.

Triangular bridge: Create a pyramid shaped supporting structure, with a sling in the middle of it. So that it could hold heavy object. The exact height is determined by the length of the straw, the longer the better.

The “Special” bridge: Crush half of the straw and place them one-on-one with the normal straw and wrap them with a lot of tape to fix them. So that the yield strength of the bridge will be stronger.

Step5: After obtaining the specific production plan, the team members formulated a plan for the evaluation of the bridge by evaluating the rationality of the six bridge design structures and the feasibility of using paper straw materials.

Step6: the evaluation of each bridge are as follows:

Arch bridge: The structure is relatively simple and thus easy to build. The fundamental basis endows its stability on the ground and the arching part supports the weight on it. However, the weight may fall from the bridge since the deck of the bridge is not level.

Suspension bridge: The capability of the maximum supporting weight is comparatively higher. And it keeps the weight safely on the deck of the bridge. However, the walls may bend or even break since this structure lacks the fundamental.

Beam bridge: The construct way is very easy, and the design is quite simple. The bridge is so flat that its load-bearing capacity is relatively poor. Because there is no pier support, the middle of the bridge is very easy to bend excessively and exceed the minimize deflection.

Truss bridge: The truss form of the bridge gives a stable form capable of supporting considerable external loads over a large span. This kind of bridge use a relatively small amount of material for weight they can support.

Triangular bridge: The only problem is the length of straw is not long, so to some extent, the supporting structure could be arched and there could be more slings.

The “Special” bridge: Creative, excellent. The shortcoming is there may not be enough straws.

Step7: Taking advantages of the six schemes above, we designed a new bridge in a trapezoidal structure. Firstly, some of the straws are folded and inserted inside another group of straws to solidify the deck. Then, the straws are piled up to 3 folds and formed the trapezoidal structure.

According to $\delta = \frac{4l^3F}{Et^4}$, The type variable of the bridge is 0.0608m. so this structure can make the bridge much stronger.